

ANALYZING THE RELATIONSHIP BETWEEN CO₂ EMISSIONS AND CLIMATE CHANGE INDICATORS IN SELECTED COUNTRIES (2000–2023)

MAT 3110 - Seminar



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Introduction

- ✔ Climate change is increasing due to human activities
- ✔ CO₂ emissions are a major contributor to global warming
- ✔ Both developed and developing countries are affected differently



Problem

Despite international climate agreements, there is limited comparative understanding of how CO₂ emissions affect temperature, sea level rise, and rainfall patterns across developed, developing, and vulnerable countries such as the USA, China, India, Germany, and Indonesia.

Objectives

- ✔ To analyze the trends in CO₂ emissions across the USA, China, India, Germany, and Indonesia from 2000 to 2023
- ✔ To evaluate the relationship between CO₂ emissions and specific climate indicators (e.g., Average Temperature, Sea Level Rise, Rainfall)
- ✔ To assess the moderate role of renewable energy adoption and forest area on these climate indicators



Significance of Study

- ✔ Helps policymakers understand climate impacts
 - ✔ Identifies vulnerable countries
 - ✔ Supports sustainable environmental planning
 - ✔ Fills existing research gaps
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Research Gap

- ④ Many studies focus on global trends
- ④ Limited country-level comparative studies
- ④ Lack of comparison between developed and vulnerable countries
- ④ Need for policy-focused findings

Literature Review

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CO₂ and Climate Change Nexus

- Rising greenhouse gases increase global warming
- Higher temperatures cause climate extremes
- Sea levels continue rising globally

Economic Growth vs Environmental Degradation

- Industrial growth increases emissions
- China → rapid industrialization
- Germany → sustainability improvements
- Environmental Kuznets Curve concept

Vulnera Role of Renewable Energybility Context

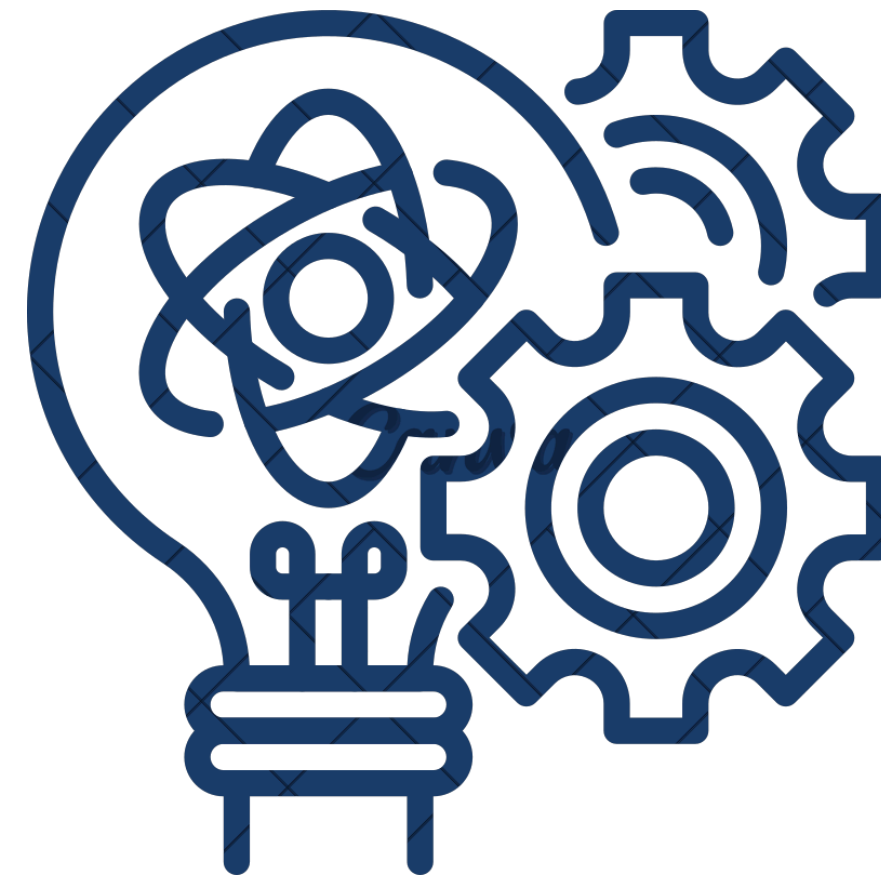
- Renewable energy reduces fossil fuel dependency
- Helps lower carbon emissions
- Green technology improves sustainability

Vulnerability Context

- Indonesia faces higher climate risks
- Sea level threats due to island vulnerability
- Agriculture and tourism impacts

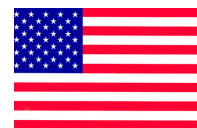
Research Design

- Quantitative research design
- Longitudinal comparative study
- Secondary panel data analysis



Data Collection

- Secondary panel data
- 2000–2023 period
- 24 observations per country
- Total observations = 120
- Countries:
 - USA
 - China
 - India



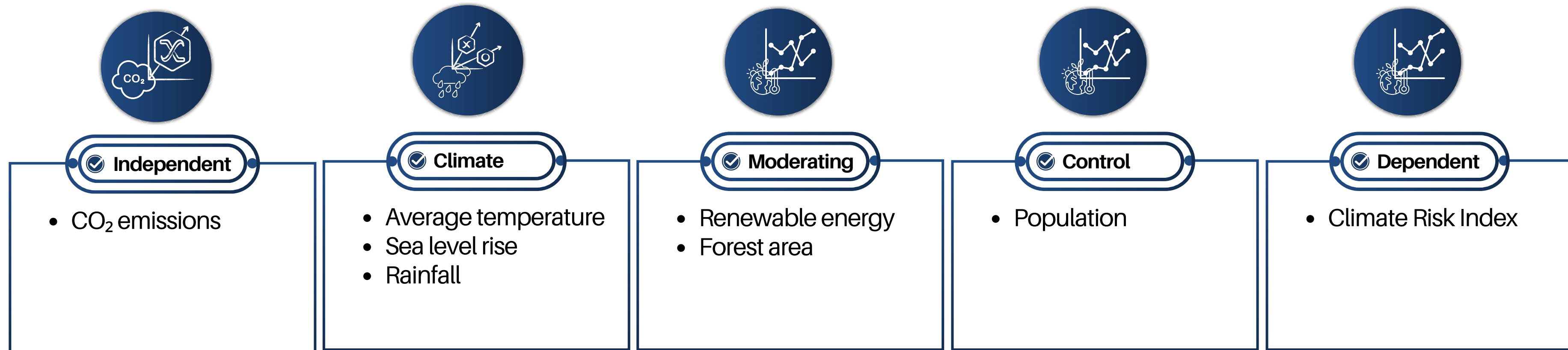
- Indonesia
- Germany



Methodology

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Variables



Mathematical Model Specification

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1

Multiple Regression Model

$$Y = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + B_4X_4 + B_5X_5 + B_6X_6 + B_7X_7 + \epsilon$$

- (Y) = Climate Change Indicator
- (X₁) = CO₂ Emissions
- (X₂) = Average Temperature
- (X₃) = Sea Level Rise
- (X₄) = Rainfall
- (X₅) = Renewable Energy
- (X₆) = Forest Area
- (X₇) = Population
- (β₀) = Intercept
- (β₁) = Regression Coefficients
- (ε) = Error Term

2 Moderation Model

Test whether Renewable Energy and Forest Area can weaken or strengthen the effect of CO₂ emissions on climate risk.

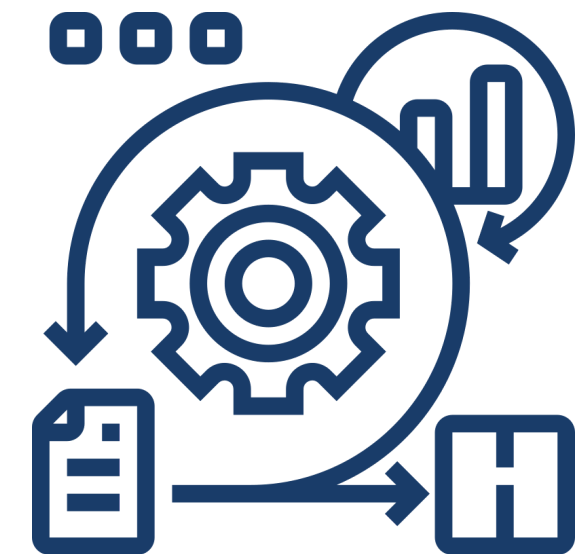
$$Y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_5 + \alpha_3 X_1 X_5 + \alpha_4 X_6 + \alpha_5 X_1 X_6 + \epsilon$$

3 Logistic Regression Model

Predict the probability of High Climate Risk

Data Analysis Techniques

- Descriptive statistics
- Reliability testing
- Normality testing
- Pearson correlation
- Multiple regression
- Moderation analysis
- ANOVA
- Independent sample t-test
- Binary logistic regression
- SPSS software



Results

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Descriptive Statistics

- Mean CO₂ emissions = 6.84
- Mean temperature = 19.58
- Mean sea level rise = 4.08
- Mean rainfall = 1675.79
- Mean renewable energy = 34.05
- Mean forest area = 37.57
- Mean climate risk = 57.23



Results

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Reliability Analysis

- Cronbach Alpha Values:
- Carbon impact → 0.970
- Temperature change → 0.976
- Sea level risk → 0.972
- Rainfall variability → 0.964
- Climate risk → 0.969

Conclusion:

All variables show
excellent reliability
(>0.90)

Normality Analysis

Variable	Skewness	Kurtosis
CO ₂	-0.068	-1.505
Temperature	0.053	-0.97
Sea level	0.158	-0.843
Rainfall	-0.112	-0.72

Conclusion:

All variables are normally distributed and suitable for parametric tests.

Results

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Correlation Results

Positive Relationships:

- CO₂ emissions → 0.739
- Temperature → 0.930
- Sea level rise → 0.990
- Rainfall → 0.764

Negative Relationships:

- Renewable energy → -0.429
- Forest area → -0.979

Interpretation:

Higher emissions increase climate risk while renewable energy and forest area reduce risk.

Regression Results

Main Regression Model

- $R = 0.991$
- $R^2 = 0.983$
- Adjusted $R^2 = 0.981$
- $F = 901.885$
- Significance = 0.000

Results

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Moderation Results

- $R^2 = 0.961$
- $F = 562.074$
- Significance = 0.000

Results

ANOVA Results

- $F = 12.584$
- Significance = 0.000

T-Test

- $t = -9.362$
- Significance = 0.000

Logistic Regression

- Chi-square = 118.648
- Accuracy = 92.5%

Discussion

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- Higher emissions increase climate risk
- Sea level rise is strongest predictor
- Forest area reduces climate risk
- Renewable energy supports mitigation
- Developed and vulnerable countries face different challenges

Discussion & Conclusion

Conclusion

- CO₂ significantly affects climate risk
- Temperature, sea level, and rainfall worsen climate vulnerability
- Climate risk is statistically predictable

Recommendations

- Increase renewable energy adoption
- Expand forest conservation
- Reduce fossil fuel dependency
- Strengthen climate policies
- Support vulnerable countries

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THANK YOU

